

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1385

Spring Gap Tap 138 kV
September 02, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1385. The Interconnection Customer proposes a 136.2 MW battery energy storage facility interconnecting at the Spring Gap Tap 138 kV substation in the Virginia area. The request was received on January 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 423 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,348,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Reach / Bluff City 138 KV	Walworth / Bluff City Substation 138 KV	107.5	397	5.5
Alligator 500-345 kV Transformer Bank	Santee Cooper / Mountaineer Wind Energy Ctr 345 KV line	107.2	896	5.6
Spindle 500/345kV Transformer Bank	Mountaineer Wind Energy Center / Santee Cooper 345 KV	101.5	1,049	9.0
Elkton D.P / Roosevelt Rd 138 KV L	Nuclear Fuels / St Louis 138 KV	100.8	364	5.0
Houston / Elkton D.P 138 KV Line	St Louis / Nuclear Fuels 138 KV line	100.8	364	5.0
Wadsworth / Roosevelt Rd 138 KV line	St Louis / Nuclear Fuels 138 KV	100.6	374	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Reach / Bluff City 138 KV	Reconductor line	\$6,612,000	30% (\$2,020,000)
Alligator 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$35,466,000	2% (\$689,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Spindle 500/345kV Transformer Bank	Replace transformer with higher-rated unit	\$36,818,000	1% (\$401,000)
Elkton D.P / Roosevelt Rd 138 KV L	Reconductor line	\$356,000	7% (\$24,000)
Houston / Elkton D.P 138 KV Line	Reconductor line	\$785,000	7% (\$53,000)
Wadsworth / Roosevelt Rd 138 KV line	Reconductor line	\$2,388,000	7% (\$161,000)
	Total allocated to project		\$3,348,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.